Chapter 2

The Heat Equation

In this text, very little prior knowledge of PDE theory and pertaining numerical methods is assumed. It seems best to begin with a relatively simple case study to illustrate basic mathematical concepts. By appealing to a model problem, i.e. the heat equation, we motivate and immediately apply the abstract theory.

2.1 Problem Formulation

In Section 1.3.1 we have derived the heat equation as the governing equation for the temperature evolution with time inside an insulated wire. Let L be the length of the wire. We consider the heat equation in the form

$$u_t = a u_{xx}, \qquad a > 0, \quad (x,t) \in \Omega_T, \tag{2.1}$$

where we have assumed for simplicity that a is constant. We are interested in the temperature at some final time $t = T$. The domain Ω_T in which we require u to satisfy the heat equation is depicted in Fig. 2.1. The scenario is informed by our derivation in section 1.3.1: Clearly, the equation must hold in the interior, i.e. $(x,t) \in \Omega := (0,L) \times (0,T)$. At time $t = 0$ we supply initial conditions, while at $x = 0$ and $x = L$ we have some kind of boundary condition. Let us assume for the moment that we prescribe the temperature there. On the other hand, at $t = T$ we expect that the temperature is determined by the time evolution of the problem. We thus define the domain $\Omega_T := (0,L) \times (0,T]$ as indicated in Fig. 2.1. The function u must satisfy the heat equation up to, and including the final time $t = T$. We denote as $\Gamma := \overline{\Omega} \backslash \Omega_T$ the part of the boundary $\partial\Omega$ where we prescribe boundary conditions. Following the simple (even somewhat simplistic) idea that we can arbitrarily prescribe a certain temperature at the end points of the wire at $x = 0$, and $x = L$, we require

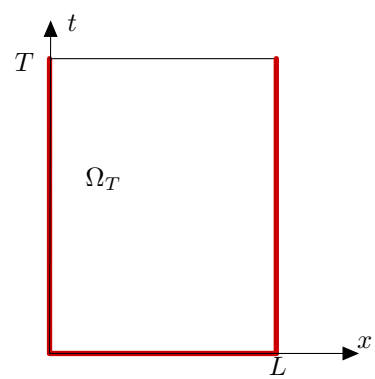

Figure 2.1: Computational domain.

$$\begin{aligned} u(0,t) &= g_1(t), \\ u(L,t) &= g_2(t), \end{aligned}$$

for $t \in [0,T]$, where g_1 and g_2 are given functions. This is called *Dirichlet* boundary condition. We define the initial conditions such that the temperature at $t = 0$ is given by some known function,

$$u(x,0) = u_0(x), \qquad x \in [0,L].$$

A subtle question is what happens at the corner points $(x,t) = (0,0)$ and $(x,t) = (0,L)$. In our setup these points are part of the domain of both the initial conditions and the boundary conditions. This ambiguity is resolved, if one notes that the physics of the problem seems to indicate that the temperature should at least be continuous on the entire boundary. Otherwise jumps in the temperature would occur, which does not seem proper. This implies that the initial conditions must also satisfy the boundary conditions. In summary, the class of functions in which we ought to look for a solution appears to be such that the unknown function u has to be differentiable sufficiently many times in Ω_T (i.e. twice with respect to x, and once with respect to t, as required by (2.1)), and has to be at least continuous up to, and including the boundary. We write this concisely as $u \in C^{2,1}(\Omega_T) \cap C(\overline{\Omega})$.

Remark 2.1 Before we proceed, let us note that Dirichlet boundary conditions are not the only kind that make sense from a physical perspective. Instead, we could also assume that the wire is perfectly insulated at the end points. This means that there is no heat flux into or out of the wire. Since the heat flux is proportional to the temperature gradient, this means that

$$\begin{aligned} u_x(0,t) &= 0, \\ u_x(L,t) &= 0. \end{aligned}$$

This is called a *Neumann* boundary condition.

From a physical perspective it seems unlikely that we will be able to control the temperature *instantly* at both endpoints, or that we will be able to insulate the wire perfectly. More elaborate boundary conditions can be derived. As an example consider the boundary condition at the end point $x = L$. We imagine that this endpoint is immersed in a reservoir of temperature $g(t)$. If we had perfect thermal conduction between the endpoint and the reservoir we would obtain the Dirichlet boundary condition $u(L,t) = g(t)$. A more realistic model is Newton's law of cooling. Here, the heat flux out of or into the wire is proportional to the temperature difference between the endpoint of the wire and the reservoir:

$$\underbrace{-au_x(L,t)}_{q} = C(u(L,t) - g(t)); \qquad C > 0.$$

For example, if $u > g(t)$, then the wire at $x = L$ is hotter than the temperature we supply, and consequently $q > 0$, which implies $u_x < 0$, and heat is flowing out of the domain. This boundary condition of mixed type is called *Robin* boundary condition. The constant C is a measure of how well we can control the temperature at the endpoint. For $C \to \infty$, we recover the Dirichlet boundary conditions. This means we can control the temperature infinitely fast. On the other hand, for $C = 0$, have Neumann boundary conditions, meaning complete insulation at the endpoint.

We begin our mathematical analysis of the heat equation with a simple model problem, restricting ourselves to $L = 1$, and homogeneous boundary conditions:

Model Problem 1 Find $u \in C^{2,1}(\Omega_T) \cap C(\overline{\Omega})$, such that

$$\begin{aligned} u_t &= au_{xx}, && (x,t) \in \Omega_T, && (2.2) \\ u(x,0) &= u_0(x), && x \in [0,1], && (2.3) \\ u(0,t) &= 0, && t \in [0,T], \\ u(1,t) &= 0, && t \in [0,T]. \end{aligned}$$

Recall that the initial conditions should be continuous, and satisfy the boundary conditions. We take $u_0 \in V$, where

$$V := \{v \in C([0,1]) : v(0) = 0, v(1) = 0\}. \tag{2.4}$$

One can easily show that a more general problem with non-homogeneous but constant Dirichlet boundary conditions can be transformed into Model Problem 1. Indeed, if v solves

$$\begin{aligned} v_t &= a v_{xx}, && (x,t) \in \Omega_T, \\ v(x,0) &= v_0(x), && x \in [0,1], \\ v(0,t) &= g_1, && t \in [0,T], \\ v(1,t) &= g_2, && t \in [0,T], \end{aligned}$$

where g_1 and g_2 are constants, then $u = v - (g_1 + x(g_2 - g_1))$ solves model problem (1) with $u_0(x) = v_0(x) - (g_1 + x(g_2 - g_1))$.

Remark 2.2 It should be stressed at this point that, in the formulation of Model Problem 1, the functional spaces have been chosen in the most general way that seemed to make sense. That the solution ought to be differentiable (twice in space, once in time), seems to be required by the PDE. That the solution ought to be continuous up to the boundary seemed to make sense from a physical perspective.

Quite often, however, one of the following two things happen: Either it turns out that the problem has no solution in the class of functions we have envisioned as appropriate (in which case we need to go back to the drawing board), or, we find out *a posteriori* that the solutions will, in fact, be even smoother than required by the most general problem formulation. In case of the heat equation, we have already in the introduction hinted at the fact, that the latter case will be true. We come back to this below.

2.2 A Maximum Principle

In this section, we shall state a very useful lemma (a preliminary result), that will facilitate the analysis of Model Problem 1 enormously.

Lemma 2.1 *Assume that a function $u \in C^{2,1}(\Omega_T) \cap C(\overline{\Omega})$ solves Model Problem 1. Then u attains both its maximum and its minimum on the boundary $\Gamma := \overline{\Omega} \backslash \Omega_T$.*

Proof We prove the assertion about the maximum. The corresponding result for the minimum can be proved in a very similar manner.

First, let M be the maximum attained by u on Γ. (Note that Γ is a closed and bounded domain, so any continuous function must attain a maximum on Γ.) Now consider arbitrary $\varepsilon > 0$, and define the function $v : \overline{\Omega} \to \mathbb{R}$, $v(x,t) = u(x,t) + \varepsilon x^2$. By assumption u solves Model Problem 1, so

$$v_t - a v_{xx} = u_t - a(u + \varepsilon x^2)_{xx} = \underbrace{u_t - a u_{xx}}_{=0} - 2\varepsilon a < 0, \qquad \forall (x,t) \in (0,1) \times (0,T]. \tag{2.5}$$

If $v(x,t)$ attains a maximum at some interior point, say at (x_0, t_0) with $0 < x_0 < 1$ and $0 < t_0 < T$, then from elementary calculus we know $v_t = 0$ and $v_{xx} \leq 0$, which contradicts (2.5).

It is easy to show that v cannot attain a maximum at $t = T$ either. Indeed, if we assume that v has a maximum at (x_0, T) for some $0 < x_0 < 1$, then we must have $v_{xx} \leq 0$, and furthermore, $v(x_0, T) \geq v(x_0, t)$, for any $0 \leq t < T$. This implies that

$$v_t(x_0, T) = \lim_{\delta \to 0^+} \frac{v(x_0, T) - v(x_0, T - \delta)}{\delta} \geq 0, \tag{2.6}$$

which again contradicts eq. (2.5). But $\overline{\Omega}$ is a closed and bounded domain, so v must attain a maximum somewhere. Since this maximum cannot be attained in Ω_T it must be attained on $\Gamma = \overline{\Omega} \backslash \Omega_T$. But from the definition of v it is clear that $v \leq M + \varepsilon$ on Γ, and so, by the preceding arguments, $v \leq M + \varepsilon$ in the entire domain $\overline{\Omega}$. The definition of v implies $u \leq M + \varepsilon(1 - x^2)$ for all $(x, t) \in \overline{\Omega}$. Recall that this is true for *any* $\varepsilon > 0$. Consequently, $u \leq M$ for all $(x, t) \in \overline{\Omega}$, and the proof is complete. ■

This version of the maximum principle is sometimes called the *weak* maximum principle. We have only proved that the values assumed by u for $(x, t) \in \Omega_T$ cannot *exceed* the maximum $M = \max_{(x,t)\in\Gamma}\{u(x, t)\}$ attained on Γ. This will be enough to serve our purposes here. There is a stronger version, which states that if $u(x, t) = M$ for some $(x, t) \in \Omega_T$, then u must be constant. (Again, similar statements hold in regard to the minimum of u.)

2.3 Well Posedness

Before solving a PDE problem, one ought to make sure that it makes sense. Naturally, mathematics cannot definitely say how close the model is to reality. Physical assumptions play a big role. But mathematics can give you some indication. For example, if somewhere along the way we have done something terribly wrong, the formulation we have derived might not make sense mathematically. Questions revolving around these thoughts are usually summarized under the paradigm of *well posedness*. We should start by giving a definition of what we mean in general when we say, a PDE problem is well posed. We formulate the definition in a somewhat "loose" language, because the quantification will depend, to some extent, on the context.

Definition 2.1 *Consider a general initial/boundary value PDE problem. We say such a problem is* well posed *if, for all admissible data,*

1. *(existence) the problem has a solution;*
2. *(uniqueness) the solution is unique;*
3. *(stability) the solution depends continuously on the data.*

The first two items in the above definition appear clear, and do not require much explanation. If the problem has no solution at all, it does not make sense to try and solve it. Likewise, if the problem admits many solutions, which one, if any, is meaningful? The third property is perhaps not so clear. First of all, what is the "data"? The data is basically everything that is given, e.g. initial conditions and boundary conditions. Small changes in those data should lead to small changes in the solution (as opposed to arbitrarily large ones). In particular, for Model Problem 1, the only given data are the initial conditions, which we had taken from the class of continuous functinos (i.e. $u_0 \in V$, where V was defined in (2.4)). The appropriate norm is the maximum norm $\|\cdot\|_\infty$, and condition 3 in Definition 2.1 can be quantified as

$$\max_{(x,t)\in\overline{\Omega}} |u(x, t)| \leq C \max_{x\in[0,1]} |u_0(x)|. \tag{2.7}$$

Note that this property follows directly from the maximum principle with $C = 1$. Or, equivalently, we can say, that if u, v are two solutions to Model Problem 1, with initial conditions

$u_0, v_0 \in V$, respectively, then, by linearity, the function $w = u - v$ is a solution with initial conditions $w_0 = u_0 - v_0$. In this case the maximum principle implies

$$\max_{(x,t)\in\overline{\Omega}} |u(x,t) - v(x,t)| \leq C \max_{x\in[0,1]} |u_0(x) - v_0(x)|. \tag{2.8}$$

This, perhaps, makes more obvious how the solution "depends continuously" on the data. Note that this implies uniqueness as well, provided solutions do, in fact exist. All that is left, then, is to show that solutions exist for any initial data $u_0 \in V$. We will now proceed to formally show that Model Problem 1 is well posed, noting that it is enough to show this for initial data in a dense subspace $V_0 \subset V$.[1]

Theorem 2.1 *Model Problem 1 is well-posed in the sense of Definition 2.1*

Proof Recall that we chose $u_0 \in V$, where V is defined in eq. (2.4). We first show that there exists a unique solution for $u_0 \in V_0$ where

$$V_0 = \left\{ v \in V : v = \sum_{k=1}^{n} c_k \sin(k\pi x),\ c_k \in \mathbb{R},\ n = 1, 2, \ldots \right\}. \tag{2.9}$$

That is, we consider the space of all functions that can be written as a *finite* Fourier sine series. It is known from Fourier theory hat this functional space is dense in V, using the norm $\|\cdot\|_\infty$, where $\|v\|_\infty = \max_{0\leq x\leq 1} |v(x)|$. For $u_0 \in V_0$ we can write for some integer n, and real numbers $c_1, \ldots, c_n$,

$$u_0 = \sum_{k=0}^{n} c_k \sin(\pi k x).$$

It is easily verified by direct differentiation that

$$u(x,t) = \sum_{k=0}^{n} c_k e^{-ak^2\pi^2 t} \sin(\pi k x),$$

satisfies the heat equation, as well as initial conditions and the boundary conditions. (Note that we do not have a problem differentiating inside the sum, because the sum is finite!) Existence of a solution is thus clear. Note that any such solution is infinitely many times differentiable. (Hence much smoother than required by the heat equation.)

We now show uniqueness. Assume that there are two solutions u_1, u_2 for the same initial conditions $u_0 \in V_0$. Then for any two such solutions, u_1 and u_2 say, the function $w := u_1 - u_2$ satisfies the same problem with $w(x, 0) = 0$. As $V_0 \subset C^{2,1}(\Omega_T) \cap C(\overline{\Omega})$, we can apply the maximum principle (Lemma 2.1), which asserts that we must have $w(x,t) = 0$ for all $(x,t) \in \overline{\Omega}$, because $w = 0$ for all $(x,t) \in \Gamma$, and consequently $u_1 = u_2$.

That the solution depends continuously on the data, we have already noted, when we established (2.8)) by the maximum principle, for two solutions $u(x,t)$ and $v(x,t)$ with initial data $u_0(x)$ and $v_0(x)$.

This shows that the problem is well-posed for initial data in V_0. The corresponding result for $u \in V$ follows by the density arguments outlined earlier in this section. ∎

[1] If the problem is well posed for $u_0 \in V_0$, there exists a unique solution, and, for any $t \in [0, T]$, we can view the solution at time t as a function of the spatial variable, given by a mapping $E(t)u_0$. Here, $E(t)$ is a linear operator, the *solution operator*, which maps the initial conditions to the (unique) solution at time t. For a well-posed problem, the solution operator must be uniformly bounded in time. Recall that any linear bounded operator defined on a dense subspace of a Banach space V can be uniquely extended to the entire Banach space (cf. Theorem A.2).

We have already pointed out that solutions to problems such as the heat equation are often smoother than the general theory requires. For the heat equation, the problem makes sense for $u \in C^{2,1}(\Omega_T) \cap C(\overline{\Omega})$, and we have seen that at least all solutions $u_0 \in V_0$ are infinitely many times differentiable. That this is true also for the general case is not trivial, but nevertheless true. We omit proof here.

It is also instructive to give a counterexample, i.e. a problem formulation that is not well-posed. This happens if we consider the solution for $t < 0$. This amounts to prescribing the temperature at some final time (here $t = 0$), and asking what the temperature distribution was at earlier times (here $t < 0$). We are thus asking, if prescribing the temperature at time $t = 0$ fixes not only the future evolution of the temperature, but also the time history. On the face of it, this seems reasonable. Indeed, one can easily check that, for example, the solution constructed in the proof of Theorem 2.1 is valid also for $t < 0$. One can even show that the solution is still unique. However, this "reverse" problem is ill-posed, as the next example shows.

Example 2.1 The solution to Model Problem 1 with $t < 0$ is *not* well posed. We show that the solution does not depend continuously on the data. Consider first the initial data $u_0(x) = 0$, which admits the unique solution $u(x,t) = 0$. Now consider *perturbed* data at $t = 0$, given by

$$u_{k,0}(x) = \frac{1}{k}\sin(\pi k x).$$

The reason we call this perturbed data, is that the amplitude of $u_{k,0}(x)$ is small for large k. The exact solution for the initial function $u_{k,0}$ is now given by

$$u_k(x,t) = \frac{1}{k}\sin(\pi k x)e^{-\pi^2 k^2 a t}.$$

We now have, e.g. for $t = -1$

$$\max_{0\le x\le 1} |u_k(x,-1) - \underbrace{u(x,-1)}_{=0}| = \max_{0\le x\le 1} |u_k(x,-1)| = \max_{0\le x\le 1} |u_{k,0}(x)|e^{a\pi^2 k^2}.$$

It is clear that we cannot bound the solution in terms of the initial conditions. This means, there is no constant C, such that (2.7) hods, because the exponential grows beyond any bound if we choose k large enough. This implies that arbitrarily small perturbations in the initial data may result in arbitrarily large changes in the solution for $t < 0$. Consequently, going "backwards" in time, the heat equation is not well posed.

2.4 Further (Optional) Reading

Evans [5] and Strauss [12] both give an introduction to the heat equation. Our proof of the weak maximum principle follows the proof in [12], Section 2.3. In that book, you will also find several extensions, such as the heat equation on the whole real line ([12] , Section 2.4), and heat equation with a source, $u_t - au_{xx} = f$ ([12], Section 3.3). Evans [5] proves the results of this chapter for the multidimensional case ([5], Section 2.3). Also, the strong maximum principle, as well as the infinite differentiability of solutions are proved there.

A whole book is devoted to the heat equation [13]. Here, you can find a discussion on more subtle issues, e.g. what happens if you omit the requirement that the solution be continuous up to the boundary.

We have only noted in passing that we can view the solution to the heat equation as being generated by a solution operator, mainly with the aim to apply the extension theorem for bounded linear operators. This can be made much more explicit by formulating the heat equation as an operator equation. Such a formulation is found in the (classic) book by Richtmyer and

Morton ([11], Section 3.1), and in the more recent book by Atkinson and Han ([1], Section 5.2). It is a much more abstract, but also more general, way of looking at the problem.